Math 261-001
Quiz 7 Solutions, October 23

1. **Final answers:**

$$\boxed{(a,b)=(1,1)}, \qquad \boxed{f(x,y,z)=x^2y+xz^2+yz+C}.$$

The components of $\vec{F}_{a,b}$ are polynomials on all of $\mathbb{R}^3$, so the field is conservative exactly when its curl is identically zero. Compute

$$\begin{aligned}\nabla\times\vec{F}_{a,b} &= \left(\frac{\partial}{\partial y}(2xz+y)-\frac{\partial}{\partial z}(bx^2+z),\ \frac{\partial}{\partial z}(2xy+az^2)-\frac{\partial}{\partial x}(2xz+y),\ \frac{\partial}{\partial x}(bx^2+z)-\frac{\partial}{\partial y}(2xy+az^2)\right)\\ &= \big(0,\ 2(a-1)z,\ 2(b-1)x\big).\end{aligned}$$

For this to vanish at every point, we must have $a=b=1$. For those values, integrate $f_x=2xy+z^2$ with respect to x:

$$f=x^2y+xz^2+g(y,z).$$

The equation $f_y=x^2+z$ gives $g_y=z$, hence $g(y,z)=yz+h(z)$. Finally, $f_z=2xz+y$ gives $h'(z)=0$. Thus $h=C$, and direct differentiation verifies

$$\nabla(x^2y+xz^2+yz+C)=(2xy+z^2,\ x^2+z,\ 2xz+y)=\vec{F}_{1,1}.$$

2. **Final answers:**

$$\boxed{\oint_C \vec{F}\cdot\vec{T}\,ds=-6}, \qquad \boxed{\oint_C \vec{F}\cdot\vec{N}\,ds=6}.$$

Write $\vec{F}=(P,Q)$, where $P=y^3-2y$ and $Q=-4x+3y^2$. The enclosed rectangle D has bounds

$$\begin{cases}0\le x\le 2,\\ 0\le y\le 1.\end{cases}$$

Because C is counterclockwise and $\vec{T}$ follows its orientation, the circulation form of Green's theorem gives

$$\begin{aligned}\oint_C \vec{F}\cdot\vec{T}\,ds &= \iint_D (Q_x-P_y)\,dA\\ &= \int_0^2\int_0^1 \left[-4-(3y^2-2)\right]dy\,dx\\ &= \int_0^2 [-2y-y^3]_0^1\,dx = \int_0^2 (-3)\,dx=-6.\end{aligned}$$

For the outward unit normal $\vec{N}$, use the flux form instead:

$$\begin{aligned}\oint_C \vec{F}\cdot\vec{N}\,ds &= \iint_D (P_x+Q_y)\,dA = \int_0^2\int_0^1 6y\,dy\,dx\\ &= \int_0^2 [3y^2]_0^1\,dx = \int_0^2 3\,dx=6.\end{aligned}$$

Circulation uses Q_x-P_y; outward flux uses the divergence P_x+Q_y.

3. **Final answer:** $\boxed{4(xy+yz+xz)}$.

First compute the curl of $\vec{G} = (z^2, x^2, y^2)$:

$$\nabla \times \vec{G} = \big(2y - 0,\ 2z - 0,\ 2x - 0\big) = (2y, 2z, 2x).$$

Apply the divergence product rule with $\vec{H} = \nabla \times \vec{G}$:

$$\nabla \cdot (f\vec{H}) = \nabla f \cdot \vec{H} + f\,\nabla \cdot \vec{H}.$$

Here $\nabla f = (2x, 2y, 2z)$, and $\nabla \cdot (\nabla \times \vec{G}) = 0$. Therefore

$$\begin{aligned}\nabla \cdot \big(f(\nabla \times \vec{G})\big) &= (2x, 2y, 2z) \cdot (2y, 2z, 2x) + f \cdot 0 \\ &= 4xy + 4yz + 4xz = 4(xy + yz + xz).\end{aligned}$$

Although a curl has zero divergence, multiplying it by a nonconstant scalar function introduces the term $\nabla f \cdot \vec{H}$.

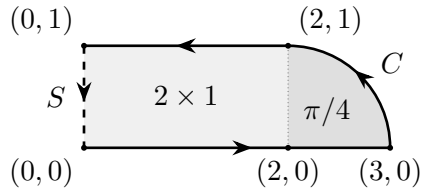
4. **Final answer:** $\boxed{5 + \dfrac{\pi}{2}}$.

The given path C is open. Add the downward segment S from $(0,1)$ to $(0,0)$; then $C + S$ is the counterclockwise boundary of a region D. For $P = ye^{xy} - y$ and $Q = xe^{xy} + x + 1$,

$$Q_x = e^{xy} + xye^{xy} + 1, \qquad P_y = e^{xy} + xye^{xy} - 1,$$
$$Q_x - P_y = 2.$$

The region is a 2×1 rectangle together with a quarter disk of radius 1 centered at $(2,0)$. Thus

$$\text{Area}(D) = 2 + \frac{\pi}{4}.$$



Green's theorem gives

$$\int_C \vec{F} \cdot d\vec{r} + \int_S \vec{F} \cdot d\vec{r} = 2\left(2 + \frac{\pi}{4}\right) = 4 + \frac{\pi}{2}.$$

On S, we have $x = 0$, $dx = 0$, and y decreases from 1 to 0. The second field component is $Q(0, y) = 1$, so

$$\int_S \vec{F} \cdot d\vec{r} = \int_1^0 1\,dy = -1.$$

Subtract this added segment's contribution to recover the original open-path integral:

$$\int_C \vec{F} \cdot d\vec{r} = \left(4 + \frac{\pi}{2}\right) - (-1) = 5 + \frac{\pi}{2}.$$